BIT 4107 – Mobile Application Development

ENHANCED PROFESSIONAL STUDENT LOGBOOK

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School/Faculty: COMPUTING AND INFORMATICS

Programme: BACHELOR OF SCIENCE INFORMATION TECHNOLOGY

Academic Year: YEAR 4

Semester: 1

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Instructions

This logbook is used to document weekly learning activities, practical exercises, skills acquired, challenges encountered, solutions applied, and reflections. Students should complete all sections every week.

Each week to follow remarks in week 1 section.

Autogenerated Table of content

Week 1: Introduction to Mobile Application Development

Practical Activities Undertaken

only screenshots needed and labeling
Screenshots/Evidence Attached (Yes/No)

Tools/Software Used

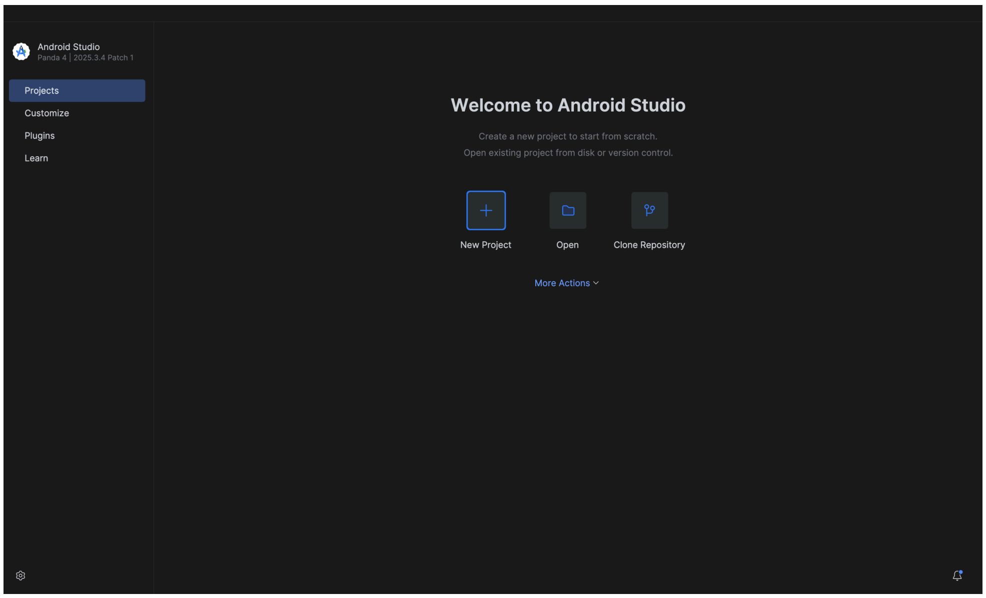


Fig 1.1: Android studio installation

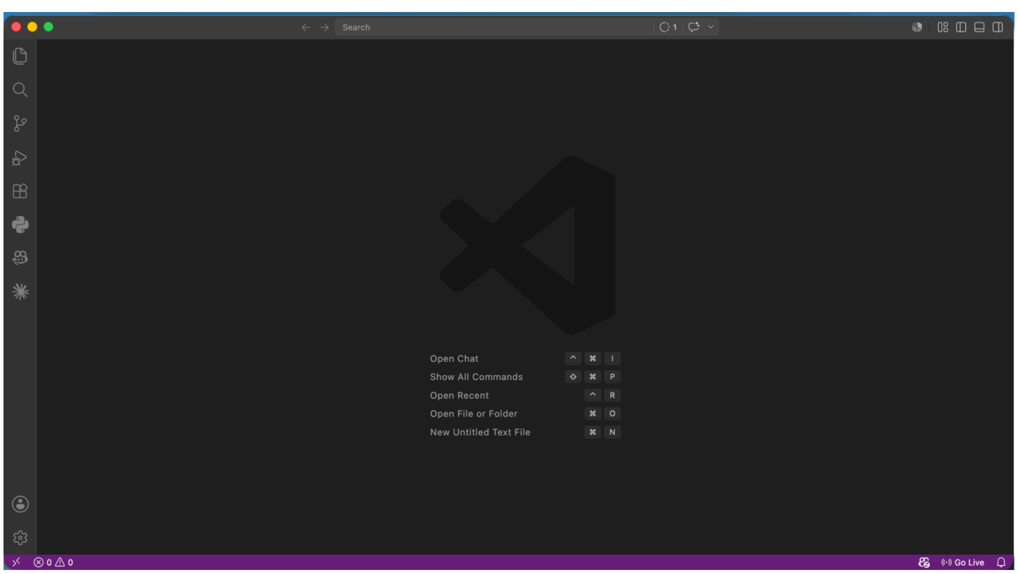


Fig 1.2: VSCode installation

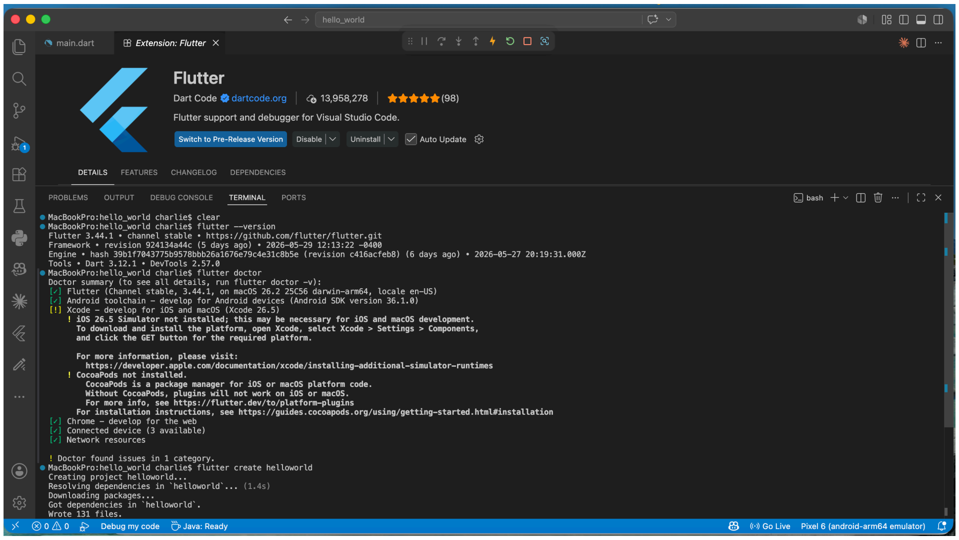


Fig 1.3: Flutter and Flutter extension installation

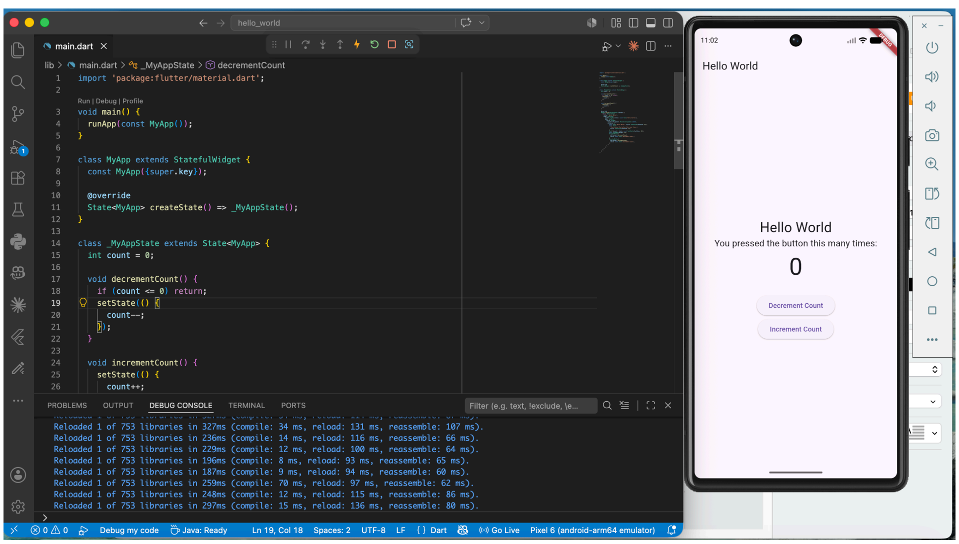


Fig 1.4: Hello world counter app

Personal Reflection

Flutter/dart appears to have a potentially good learning curve based on previous programming knowledge.

Week 2: Mobile Development Environment

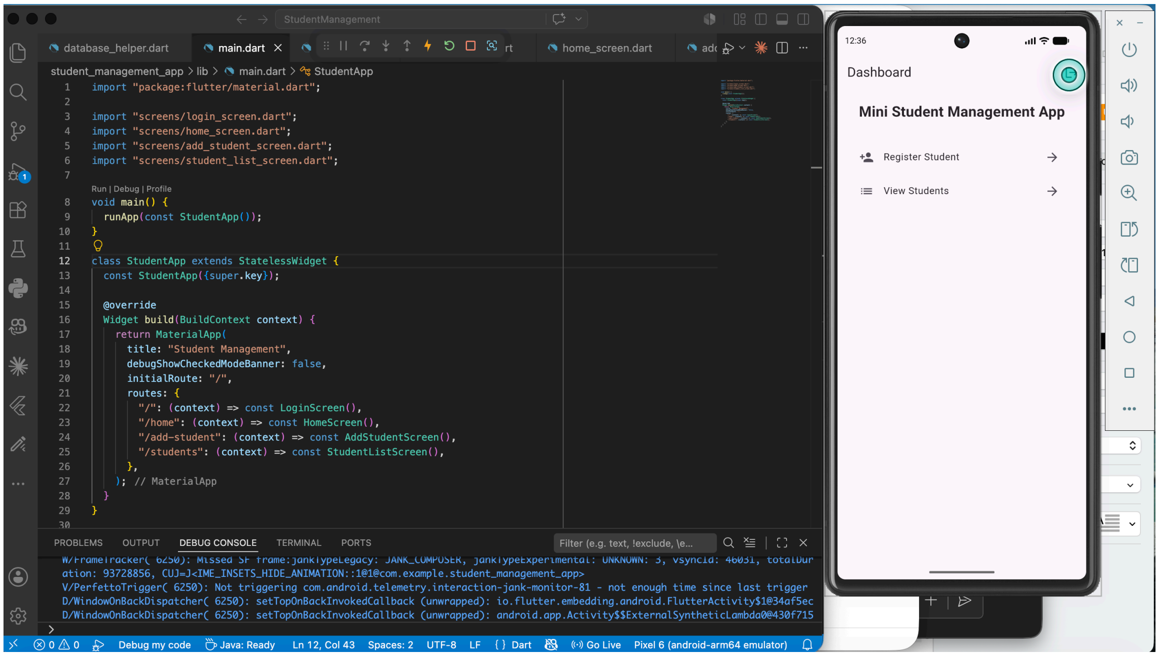


Fig 2.1: Mini student management app

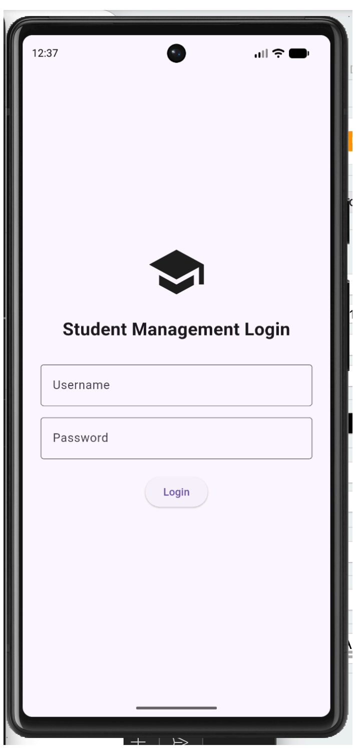


Fig 2.2: Admin Login page

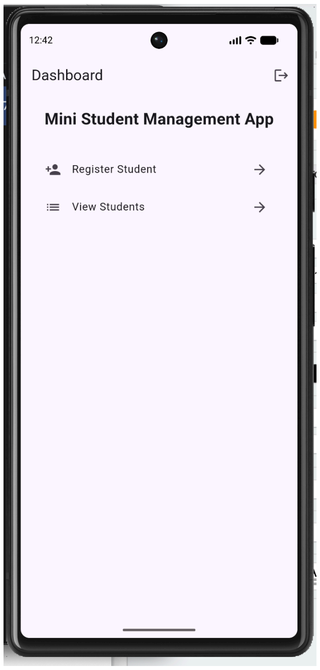


Fig 2.3: Homepage

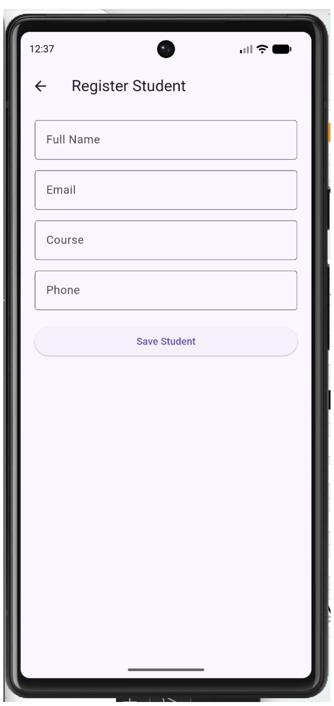


Fig 2.4: Register student

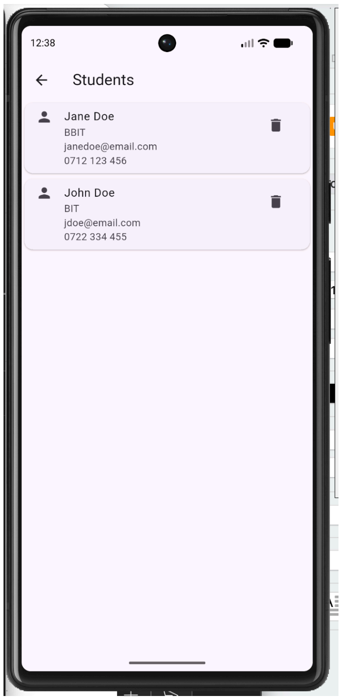


Fig 2.5: List students

Personal Reflection

Interesting insights on persisting data permanently on a local database.

Week 3: User Interface Design

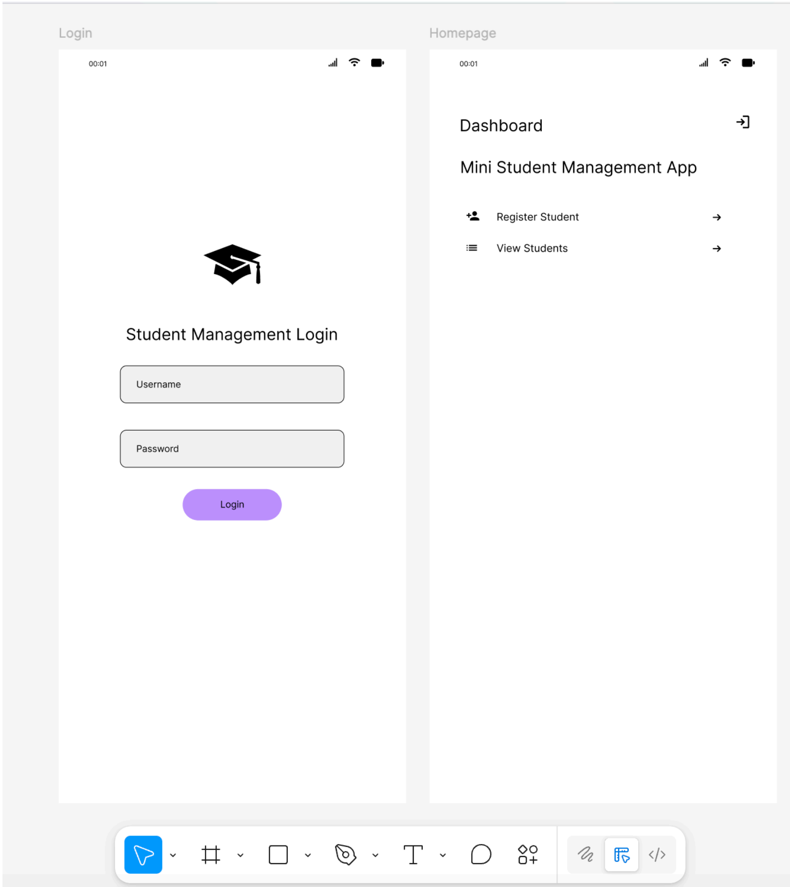


Fig 3.1: Figma - Login and Homepage wireframes

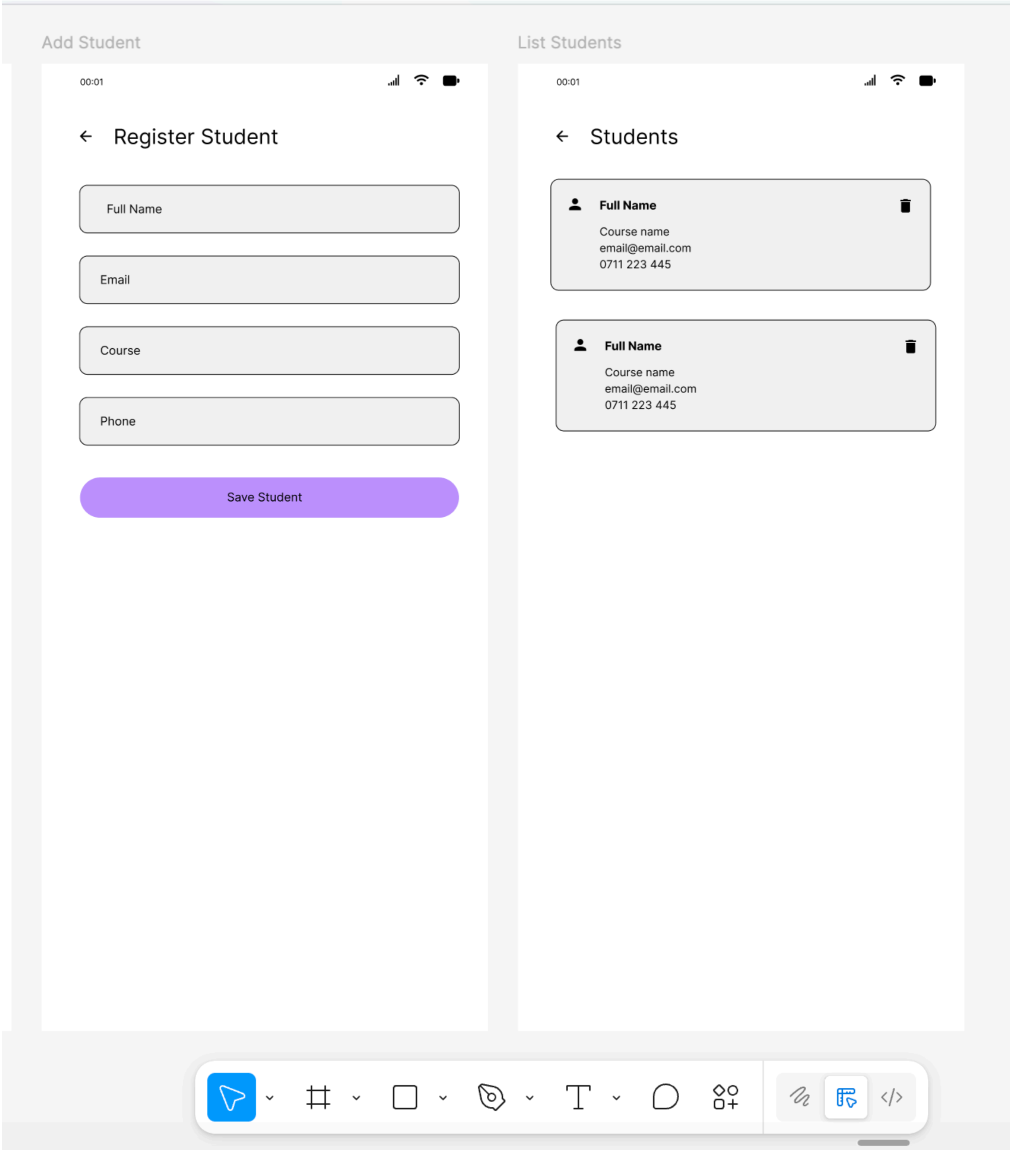


Fig 3.2: Figma - Register and List Students wireframes

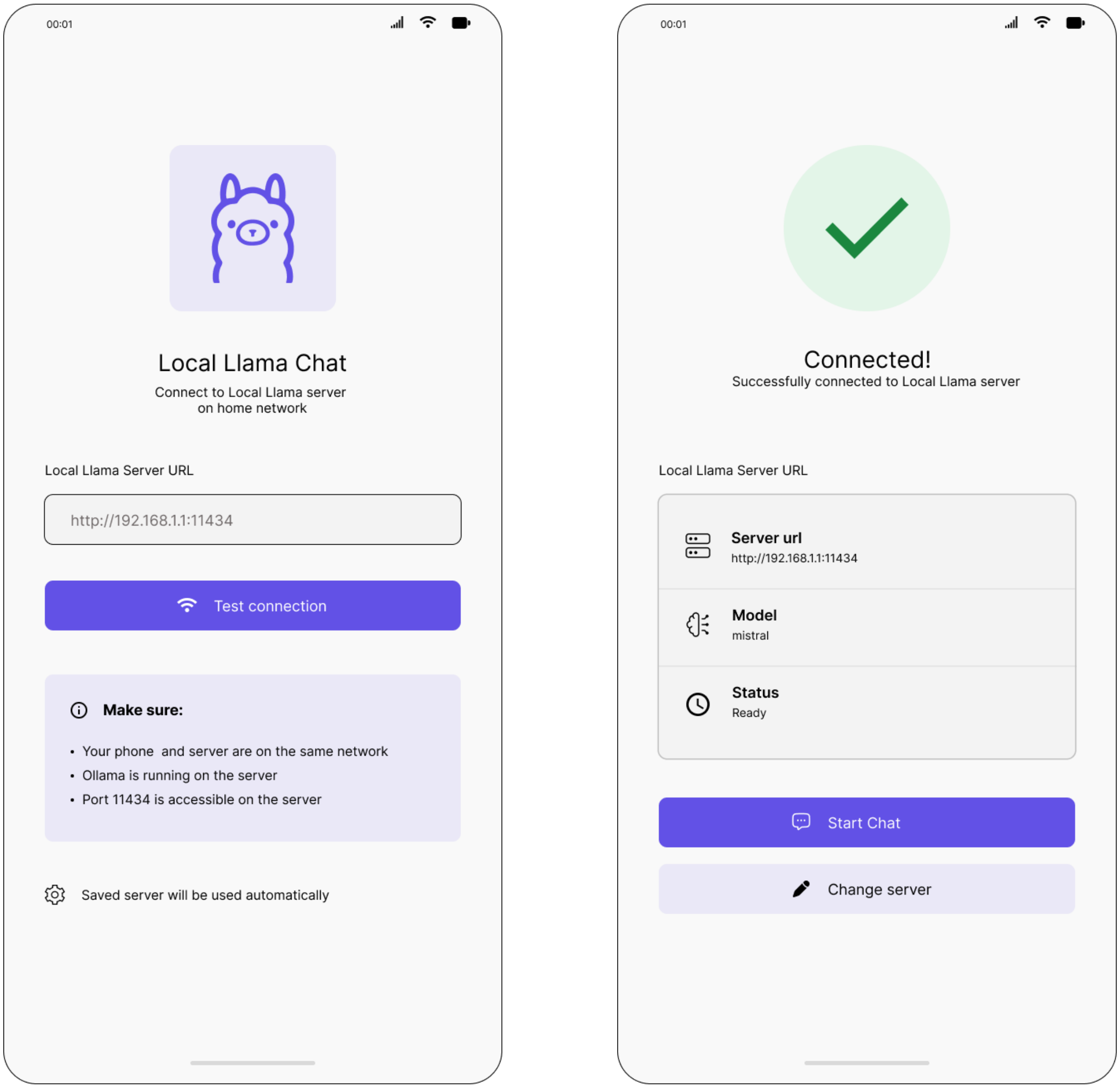


Fig 3.3: Figma - Local Llama Homepage Wireframe

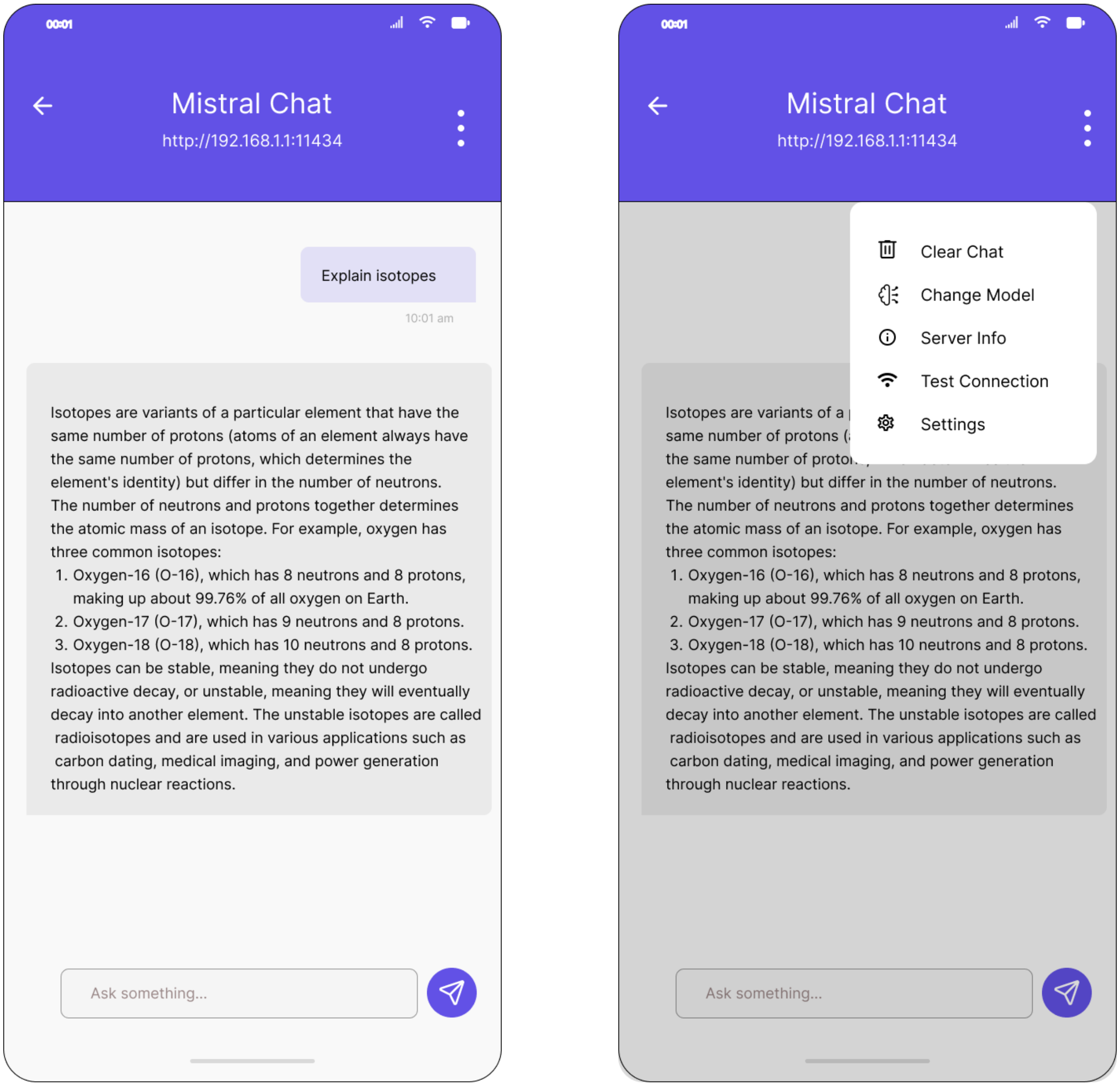
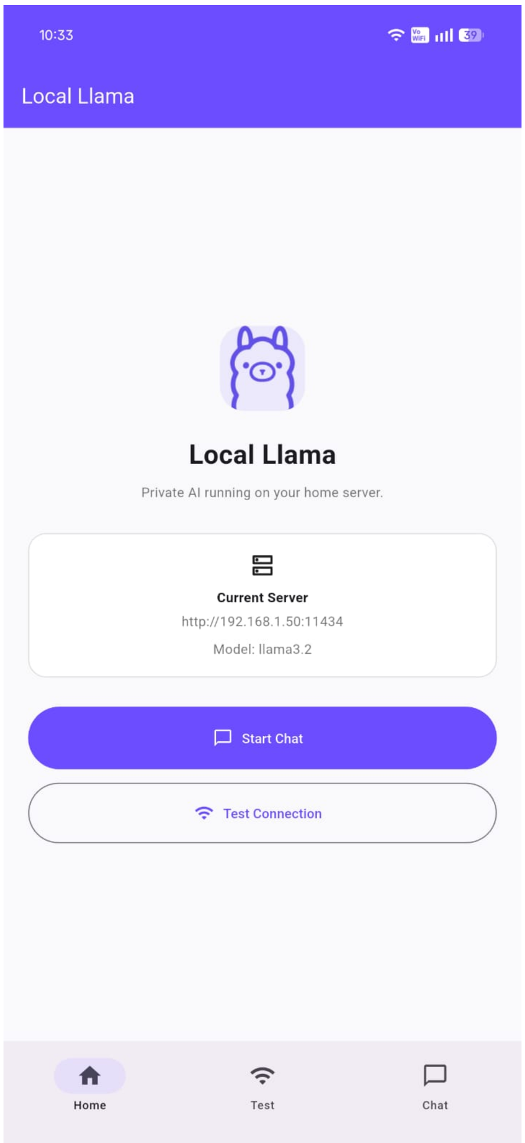


Fig 3.3: Figma - Local Llama Chat Wireframes

Personal Reflection

Prototyping first will speed up development.

Week 4: Event Handling and User Interaction



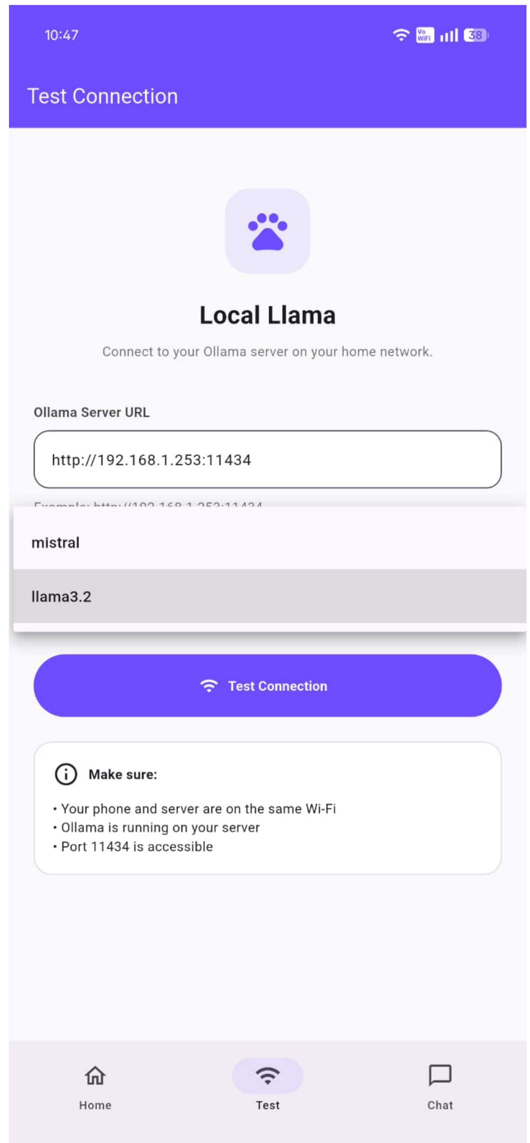


Fig 4.1:Live app - Intuitive navigation

Fig 4.2:AI Model selection

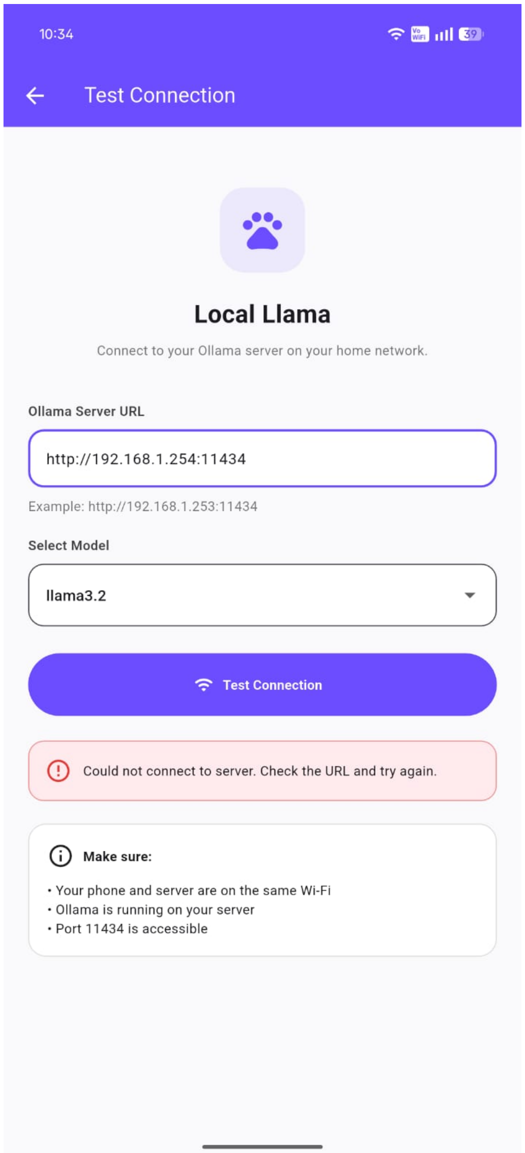


Fig 4.3:Connection event failure

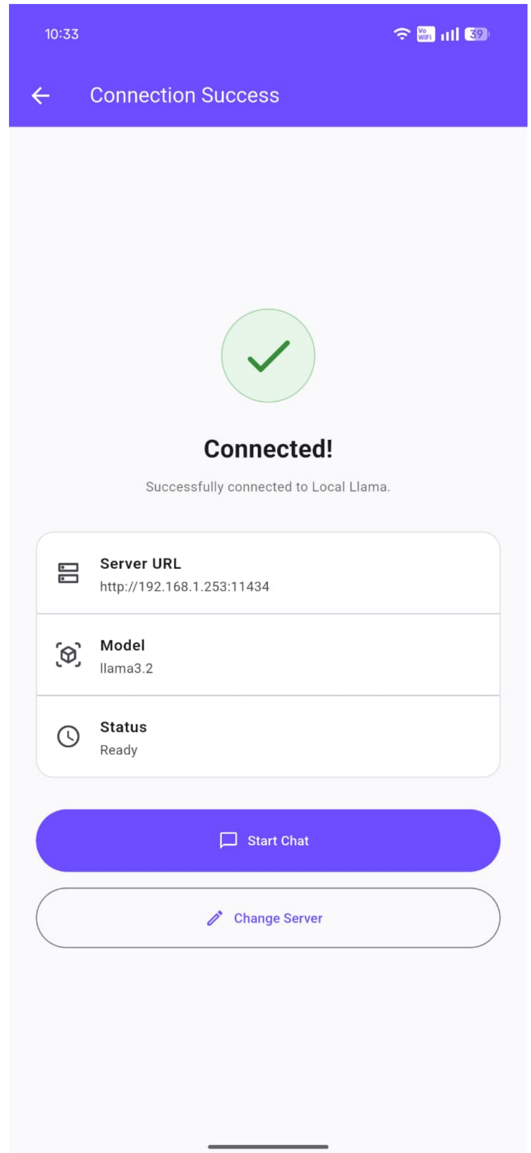
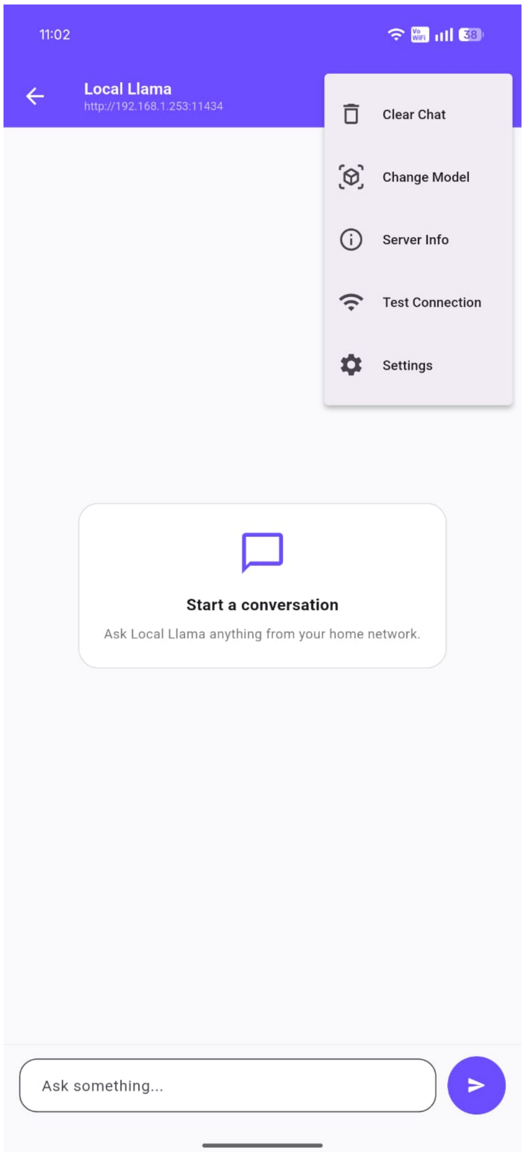


Fig 4.4: Connection event success

Week 5: Data Management

Same as week 1



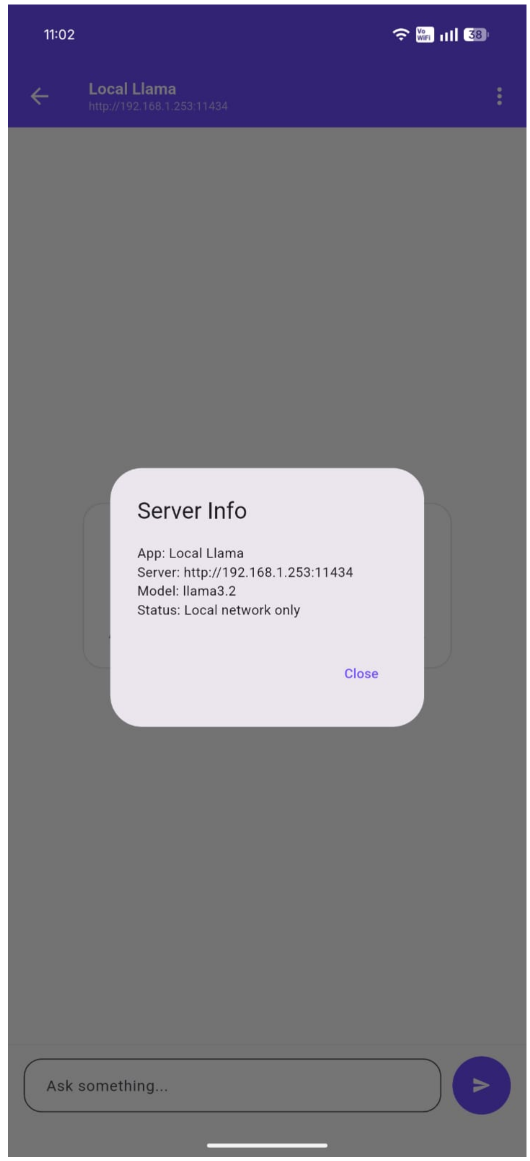
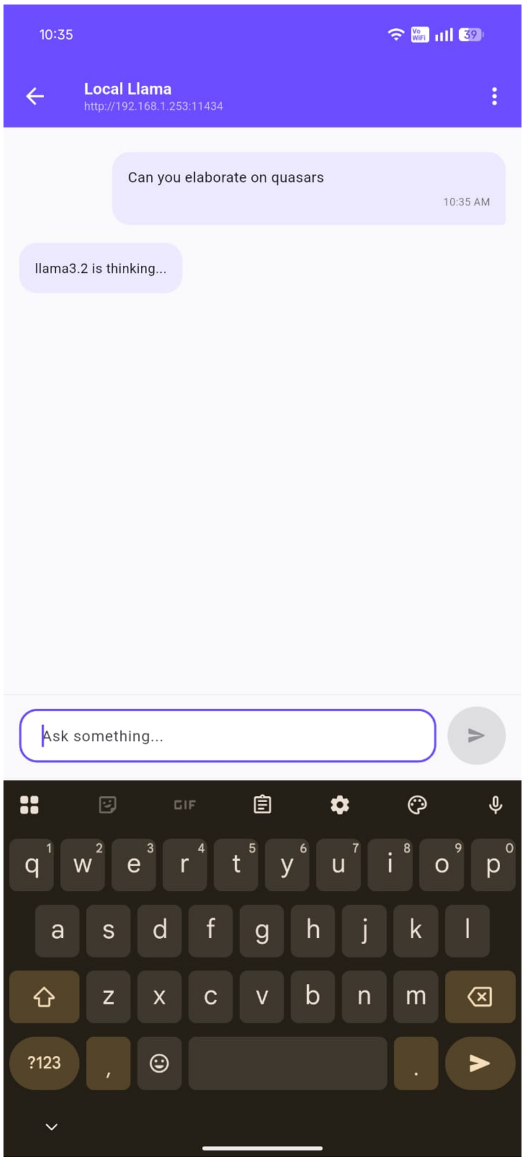


Fig 5.1:App settings data *Fig 5.2: Data in shared preferences*

Week 6: Networking and APIs



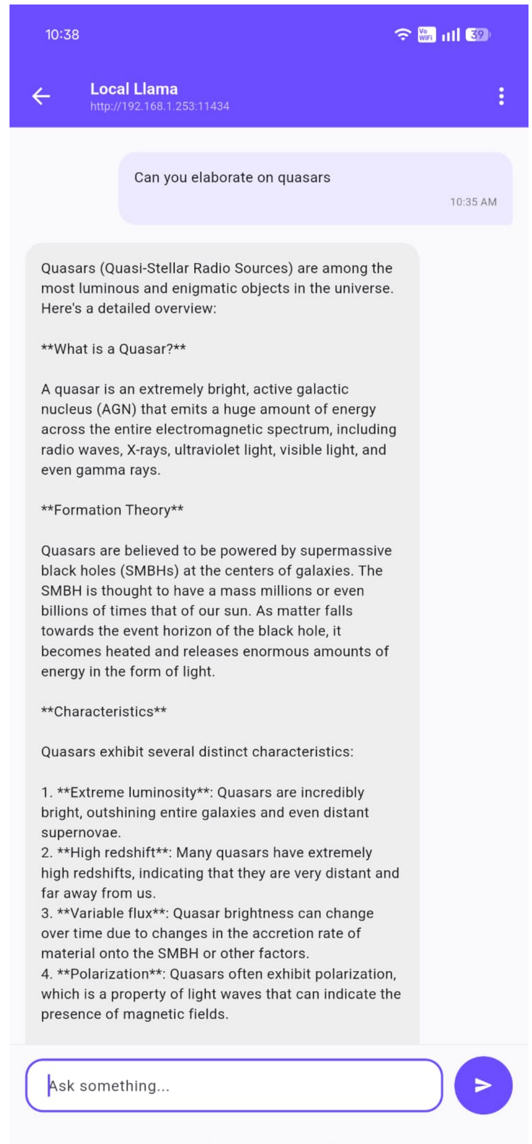


Fig 6.1:POST data to server *Fig 6.2: Data received from server*

Week 7: Application Architecture

database_helper.dart

student_management_app > lib > db > database_helper.dart > ...

```dart
class DatabaseHelper {

  Future<Database> _initDB(String filePath) async {
    final dbPath = await getDatabasesPath();
    final path = join(dbPath, filePath);

    return await openDatabase(path, version: 1, onCreate: _createDB);
  }

  Future<void> _createDB(Database db, int version) async {
    await db.execute("""
      CREATE TABLE students (
        id INTEGER PRIMARY KEY AUTOINCREMENT,
        name TEXT NOT NULL,
        email TEXT NOT NULL,
        course TEXT NOT NULL,
        phone TEXT NOT NULL
      )
    """);
  }

  Future<int> insertStudent(Student student) async {
    final db = await instance.database;
    return await db.insert("students", student.toMap());
  }

  Future<List<Student>> getStudents() async {
    final db = await instance.database;

    final result = await db.query("students", orderBy: "id DESC");

    return result.map((map) => Student.fromMap(map)).toList();
  }

  Future<int> deleteStudent(int id) async {
    final db = await instance.database;

```

Fig 7.1: SQLite Database setup

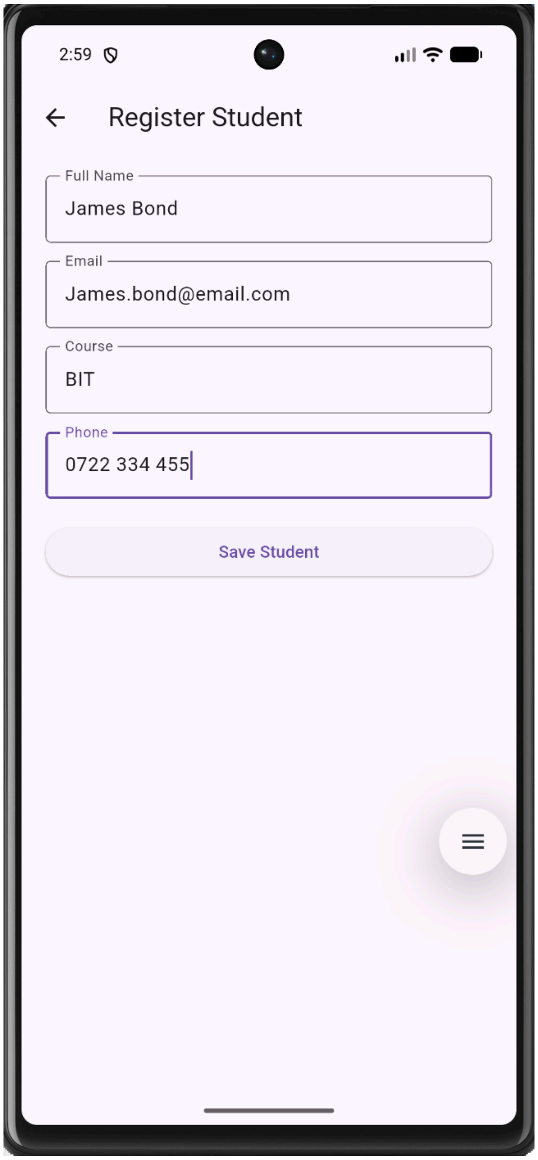


Fig 7.2: Database Add entry

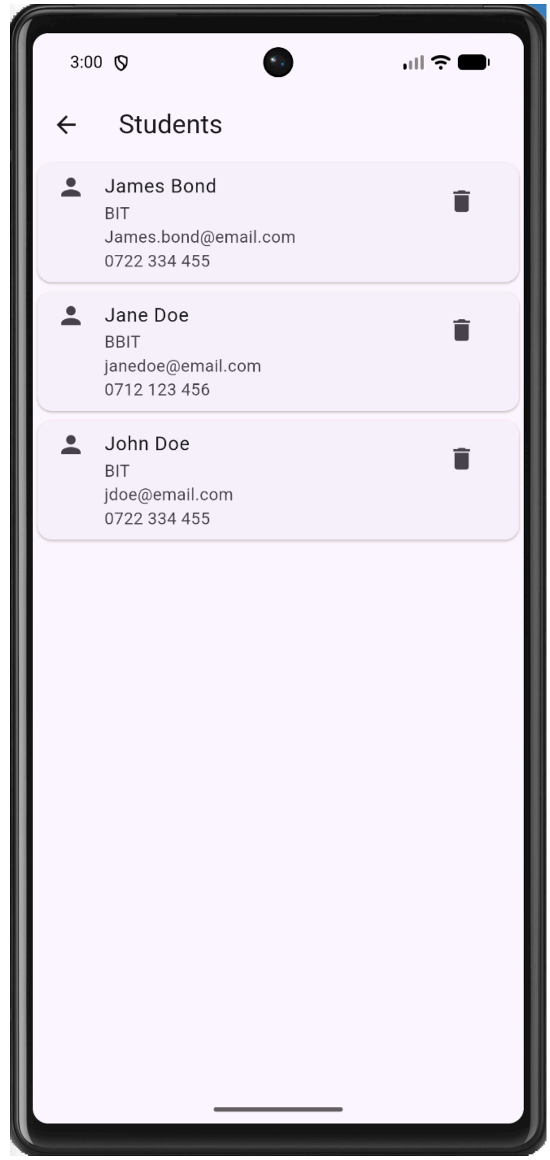


Fig 7.2: Database List rows

Week 8: Project Planning

Week 9: Device Features and Sensors

Same as week 1

Week 10: Notifications and Background Services

Same as week 1

Week 11: Mobile Security

Same as week 1

Week 12: Testing and Debugging

Same as week 1

Week 13: Application Deployment

Same as week 1

Week 14: Final Project Presentation

Same as week 1

Semester Project Tracking

Project Title

Project Objectives

Project Requirements

Development Milestones

Testing Results

Lessons Learned

Future Improvements

End of Semester Reflection

Summary of Skills Acquired

Most Interesting Topic

Major Challenges Faced

How Challenges Were Overcome

Professional Growth Achieved

Overall Course Experience

Final Approval

Student Name: ______________________

Student Signature: __________________

Date: ____________________________

Lecturer Name: _____________________

Lecturer Remarks: _________________

Date: ____________________________